A Wasserstein lower bound for local group-product networks

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Abstract

Consider a network computing a Boolean predicate of an ordered product of n letters in a fixed finite group. Suppose its physical layers merge contiguous subwords through successive local modules, with disjoint data channels between different modules in each layer band. Internal states may retain the entire subword, have arbitrary dimension and precision, and use arbitrary fixed-seed estimator features. No internal margin or product-state encoding is assumed. We prove that the product of the layer Frobenius norms is at least a constant times the exponential of one half the accumulated entropy of the subword partitions. If every merge has arity at most b, this is at least $\exp((\log n)^2/(4\log b) + (\log n)/4)$, up to the stated input and margin factor, for unbalanced as well as balanced hierarchies. The proof uses Wasserstein distances between state distributions conditioned on the group product. The lower bound concerns this physical architecture. It also applies to local parity networks, although parity has a shallow polynomial-product realization. It is therefore neither an intrinsic function-norm lower bound nor a Bayesian sample lower bound.

1 Purpose and the physical restriction

A bounded-width branching program can multiply letters from a fixed finite group in linear time. A balanced arithmetic formula can do the same with polynomial size and logarithmic depth. These observations do not by themselves supply a tanh network with small product of layer Frobenius norms: a physical network must transmit its intermediate states through its actual matrices. The issue is especially visible in local multiplication trees, which repeatedly maintain many independent subword computations in parallel.

One possible response is to change the state code. A module might use a high-dimensional code, keep auxiliary information about the original subword, or store random-estimator values whose mean represents its group product. The theorem below permits all such deterministic boundary states. Its restriction is on dataflow: before their designated merge, different subwords have disjoint data-dependent channels. Each merge band consists of at least one actual matrix layer, and every layer that carries the states is charged.

This distinction matters. A free relabeling of two vectors as one concatenated vector is not an additional physical layer. Nor may one factor a global linear map virtually, assign the factors to an invented tree, and charge the resulting logical depth. Conversely, arbitrary depth *inside* a genuine local band is permitted. The lower bound will not depend on that depth.

1.1 Inputs and target

Fix a finite group G and $n \geq 2$. The input letters $g_1, \ldots, g_n$ range independently over G. Letter i is supplied through a fixed encoding $E_i : G \to \mathbb{R}^{s_i}$. The principal case is an injective Boolean

encoding, $E_i(G) \subseteq \{-1,1\}^{s_i}$. A single augmented constant coordinate may be provided in addition, as in the usual bias-free tanh architecture.

Fix a nonconstant predicate $f : G \to \{-1,1\}$, and put

$$p = \frac{|\{g \in G : f(g) = -1\}|}{|G|} \in (0,1).$$

The network output z is required to satisfy

$$f(g_1 \cdots g_n) z(g_1, \ldots, g_n) \geq 1 \qquad \text{for every } (g_1, \ldots, g_n) \in G^n. \tag{1}$$

The output magnitude may depend on the whole word. Only its required sign depends on the group product. If the network is defined on a larger Boolean cube, (1) is needed only on valid encoded words. In particular, any full-cube extension satisfying the margin is covered.

Uniform letters are a device in the proof, used to define conditional state distributions. The theorem is a deterministic representation statement, not a learning guarantee for a specified sampling law.

1.2 Partitions, bands, and arbitrary boundary codes

Let

$$\mathcal{P}_0, \mathcal{P}_1, \ldots, \mathcal{P}_T$$

be successive partitions of $\{1, \ldots, n\}$ into contiguous intervals. The first partition consists of singletons, the last consists of the whole word, and each partition coarsens the preceding one. A block in $\mathcal{P}_j$ is a parent; the consecutive blocks of $\mathcal{P}_{j-1}$ that it contains are its children. Unary carry modules are allowed.

At the cut associated with a block v, let $e_v(w_v) \in \mathbb{R}^{r_v}$ denote its boundary code, where w_v is the complete ordered subword in that block. It is an arbitrary deterministic function of that subword. It may retain every original letter and arbitrary auxiliary values. There is no assumption that it is a function only of the product of the subword. The initial singleton codes are the prescribed encodings E_i.

The bands, in order, partition all actual matrix layers preceding the final readout. Each band is nonempty and consists of consecutive layers; no physical layer is reused or omitted. Within band j, different parent modules have disjoint data-dependent channels. The first matrix of each parent module receives only the boundary codes of its children; subsequent matrices receive only that module's data-dependent channels. Shared input-independent constants are allowed. They may be treated as biases when deriving a lower bound, although their actual weights are included in the network's norm.

All activations are coordinatewise 1-Lipschitz. They may differ between neurons and layers. Tanh is a special case; zero preservation is not needed for the proof. The final readout is linear,

$$z = v^\top e_{\text{root}} + c.$$

Its constant can be supplied through ordinary charged constant channels. Write

$$P_{\text{F}} = \prod_{\ell} \|W_\ell\|_{\text{F}}$$

for the product over all actual weight matrices, including the readout. Any additional charged constant weights only increase the lower estimates used below. There are no uncharged input-dependent bypasses.

The proof follows the physical cuts just defined. At each cut we measure how distinguishable the code distributions are when the subword product changes. A coupling bounds that signal through any one child, the actual matrix norms combine these bounds across a band, and weighted arithmetic–geometric mean makes the intermediate signal scales cancel. Only the partition entropies, input scale, and final margin remain.

2 The entropy-area theorem

Give every block u mass

$$\mu_u = |u|/n.$$

At each cut these masses form a probability vector. Define its entropy in natural units by

$$\mathcal{H}(\mathcal{P}_j) = -\sum_{u\in\mathcal{P}_j} \mu_u \log \mu_u.$$

For the input encodings, set

$$a_i^2 = \mathbb{E}_{g\sim\mathrm{Unif}(G)} \left\| E_i(g) - \mathbb{E}E_i(g) \right\|_2^2, \qquad A_0 = \prod_{i=1}^{n} a_i^{1/n}.$$

Whenever (1) is realizable in the stated architecture, all a_i are positive, as will follow from the proof. For Boolean s_i-bit encodings, $a_i \le \sqrt{s_i}$.

Theorem 2.1 (Entropy accumulated across physical bands)**.** *Every network satisfying the preceding architecture assumptions and* (1) *obeys*

$$P_{\mathrm{F}} \ge \frac{2\sqrt{p(1-p)}}{A_0} \exp\left(\frac{1}{2} \sum_{j=1}^{T} \mathcal{H}(\mathcal{P}_{j-1}) \right). \tag{2}$$

Internal dimensions, code precision, retained history, and module depths are unrestricted. Every term in the entropy sum corresponds to a nonempty band of actual charged matrix layers.

The theorem separates two effects. Changes in the numerical scale of a boundary code will telescope out. The entropy of the number and sizes of simultaneously separate subword computations remains. The next corollary makes this quantitative without requiring a balanced tree.

Corollary 2.2 (Bounded merge arity)**.** *Suppose every parent has at most b children, where $2 \le b \le n$. Then*

$$P_{\mathrm{F}} \ge \frac{2\sqrt{p(1-p)}}{A_0} \exp\left(\frac{(\log n)^2}{4 \log b} + \frac{\log n}{4} \right). \tag{3}$$

For a complete b-ary tree with $n = b^h$, this is

$$P_{\mathrm{F}} \ge \frac{2\sqrt{p(1-p)}}{A_0}\, b^{h(h+1)/4}.$$

For fixed arity the product is at least $\exp(\Omega((\log n)^2))$. For arity of order $\log n$, the corresponding lower bound is

$$\exp\!\big(\Omega((\log n)^2/\log\log n)\big).$$

When $b = n$ and there is only one band, the bound is only a constant times $\sqrt{n}$. That case does allow an arbitrary global network inside the one root band; no repeated local hierarchy is being imposed.

3 Conditional Wasserstein signals

For two finitely supported probability laws ν, ν' on a Euclidean space, their quadratic Wasserstein distance is

$$W_2(\nu,\nu')^2 = \min_{\pi\in\Pi(\nu,\nu')} \int \|x-y\|_2^2 \, d\pi(x,y),$$

where $\Pi(\nu,\nu')$ is the set of couplings with the stated marginals. The minimum exists. We use two elementary facts: applying an L-Lipschitz map to both sides multiplies this squared distance by at most L^2; and coupling mixtures with the same mixing variable bounds their squared distance by the average squared distances of the coupled components. Both facts follow by constructing the indicated coupling.

For a block u and $g \in G$, let $\nu_{u,g}$ be the law of its boundary code under uniform subwords conditioned on their product being g. Every group product has positive probability, and the code laws have finite support. Define

$$a_u^2 = \frac{1}{2|G|^2} \sum_{g,h\in G} W_2(\nu_{u,g},\nu_{u,h})^2. \tag{4}$$

The same notation agrees with the earlier input definition: at a leaf, $\nu_{i,g}$ is the point mass at $E_i(g)$, and the average squared distance of two independent points, divided by two, is their variance.

The conditional laws couple away arbitrary variation within a fixed product class. No decoding map from code to product is needed. This is the reason the argument permits complete histories and auxiliary estimator values.

Lemma 3.1 (One-child contraction)**.** *Suppose a parent module has code map $F(e_1(w_1),\dots,e_m(w_m))$. If its Lipschitz constant when only child i's code varies is at most L_i, then*

$$a_{\text{parent}} \le L_i a_i.$$

Proof. We first give the binary case, varying the left child. Conditional on the parent product being a, its complete right subword remains unconditionally uniform. Indeed the independent left product is uniform on G, so each fixed right subword contributes the same factor $1/|G|$ to the conditioning probability. If the right subword has product h, the left product must be ah^{-1}.

To couple parent code laws for products a and b, choose the same uniform right subword in both copies. Conditional on that word, couple the two left codes optimally under

$$\nu_{\text{left},ah^{-1}}, \qquad \nu_{\text{left},bh^{-1}}.$$

Apply F to these codes and the shared right code. This constructs a coupling with the desired parent marginals and gives

$$W_2(\nu_{\text{parent},a},\nu_{\text{parent},b})^2 \le L_i^2 \mathbb{E}_{w_{\text{right}}} W_2(\nu_{\text{left},ah^{-1}},\nu_{\text{left},bh^{-1}})^2.$$

For every fixed h, right translation sends a uniform pair (a,b) to a uniform pair (ah^{-1},bh^{-1}). Average the last inequality over a,b and use (4).

For an arbitrary child of an m-ary parent, condition on all the other full subwords and denote the products before and after that child by L and R. The other subwords again remain jointly uniform under parent-product conditioning, because the missing child's product is independent uniform. Its required product is $L^{-1}aR^{-1}$. The same construction and the bijection $a \mapsto L^{-1}aR^{-1}$ prove the result. □

Remark 3.2. The parent code need not be constant on product classes, and its two conditional laws need not have the same covariance or mean. The coupling uses the full conditional laws. Replacing them by the means alone would not justify the lemma for history-retaining states.

Lemma 3.3 (The output margin supplies a signal)**.** *The root signal is positive, and*

$$\|v\|_2 \geq \frac{2\sqrt{p(1-p)}}{a_{\text{root}}}.$$

Proof. For opposite labels $f(g) \neq f(h)$, every code in the support of $\nu_{\text{root},g}$ and every code in the support of $\nu_{\text{root},h}$ have readout values separated by at least two. Their Euclidean separation is therefore at least $2/\|v\|_2$. Every coupling has at least this squared displacement, so

$$W_2(\nu_{\text{root},g}, \nu_{\text{root},h})^2 \geq 4/\|v\|_2^2 .$$

The fraction of ordered opposite-label pairs is $2p(1-p)$. The factor $1/2$ in (4) gives $a_{\text{root}}^2 \geq 4p(1-p)/\|v\|_2^2$. □

Lemma 3.1 propagates positivity from the root to every preceding cut and every leaf. All signal ratios below are therefore legitimate. The readout can alternatively be included in the last root band; then its scalar output has signal at least $2\sqrt{p(1-p)}$.

4 Physical norms and the entropy telescope

Fix one band. For parent module u, partition its first variable-input matrix into blocks $B_{u,i}$, one for each child. Let T_u be the product of the Frobenius norms of its remaining variable-channel matrices, with the empty product equal to one. Coordinatewise activation contractions and the operator-norm bound by Frobenius norm give a one-child Lipschitz constant at most $T_u \|B_{u,i}\|_{\text{F}}$. Consequently

$$P_u^2 \geq T_u^2 \sum_{i \text{ child of } u} \|B_{u,i}\|_{\text{F}}^2 \geq \sum_{i \text{ child of } u} (a_u/a_i)^2, \tag{5}$$

where P_u is the product of the module's variable-channel Frobenius norms. Any constants omitted from this estimate can only increase actual matrix norms. Shared input-independent channels may be frozen as biases when bounding changes in the variable inputs.

At physical layer t of the band, channel disjointness implies

$$\|W_t\|_{\text{F}}^2 \geq \sum_u \|B_{u,t}\|_{\text{F}}^2 .$$

Expanding a product of sums of nonnegative numbers gives

$$P_{\text{band}}^2 \geq \prod_t \sum_u \|B_{u,t}\|_{\text{F}}^2 \geq \sum_u \prod_t \|B_{u,t}\|_{\text{F}}^2 = \sum_u P_u^2 \geq \sum_{i \to u} (a_u/a_i)^2. \tag{6}$$

This is why arbitrary additional physical depth inside a local band does not weaken the conclusion. The estimate only requires the band to be nonempty and all its variable-channel matrices to be charged.

Proof of Theorem 2.1. Apply weighted arithmetic–geometric mean to the last sum in (6), using weights μ_i on the child edges. These weights sum to one because the child blocks form the pre-band partition. Thus

$$\begin{aligned}\log P_{\text{band}} &\geq \frac{1}{2}\sum_{i\to u}\mu_i \log\frac{(a_u/a_i)^2}{\mu_i}\\ &= \sum_{i\to u}\mu_i(\log a_u - \log a_i) + \frac{1}{2}\mathcal{H}(\mathcal{P}_{\text{pre}}).\end{aligned}$$

For each parent, the child masses sum to its own mass. Hence, writing $A_j = \prod_{u\in\mathcal{P}_j} a_u^{\mu_u}$, the first term is exactly $\log A_j - \log A_{j-1}$. Sum over bands. The result is

$$\prod_j P_{\text{band},j} \geq \frac{a_{\text{root}}}{A_0}\exp\left(\frac{1}{2}\sum_j \mathcal{H}(\mathcal{P}_{j-1})\right).$$

Multiply by the readout bound in Lemma 3.3. All internal signal scales cancel, proving (2). □

Proof of Corollary 2.2. The entropy chain rule for one coarsening gives

$$\mathcal{H}(\mathcal{P}_{j-1}) - \mathcal{H}(\mathcal{P}_j) = \sum_{u\in\mathcal{P}_j}\mu_u\mathcal{H}\big((\mu_i/\mu_u)_{i\text{ child of }u}\big) \leq \log b.$$

Initial entropy is $A = \log n$ and final entropy is zero. With $B = \log b$ and $r = \lceil A/B\rceil$, the sum of pre-band entropies is at least

$$\sum_{j=0}^{r-1}(A - jB) = rA - \frac{Br(r-1)}{2} \geq \frac{A^2}{2B} + \frac{A}{2}.$$

For the last inequality, set $r = A/B + \theta$, $0 \leq \theta < 1$; the difference is $B\theta(1-\theta)/2 \geq 0$. Substitute this estimate into (2). When $n = b^h$ and the tree is balanced, the cut entropies are $h\log b, (h-1)\log b, \ldots, \log b$, giving the displayed exact exponent. □

Corollary 4.1 (Large interaction blocks must appear early)**.** *Fix G, f and bounded-size Boolean leaf encodings. Suppose networks in this physical hierarchy satisfy $P_{\text{F}} \leq C_0 n^K$ for fixed $C_0 > 0$ and $K \geq 0$. For every fixed $0 < \varepsilon < 1$, a block containing at least $n^{1-\varepsilon}$ letters must appear within $O((K+1)/\varepsilon)$ charged bands.*

Proof. Theorem 2.1 gives $\sum_j \mathcal{H}(\mathcal{P}_{j-1}) \leq 2K\log n + O(1)$. Thus only $O((K+1)/\varepsilon)$ pre-band cuts can have entropy at least $\varepsilon\log n$. If the hierarchy terminates earlier, its root already has the required size. Otherwise some cut within that many bands has entropy at most $\varepsilon\log n$. For any probability vector, $\mathcal{H}(\mu) \geq \log(1/\max\mu)$, so that cut has a block of mass at least $n^{-\varepsilon}$. □

The conclusion concerns this hierarchy of physical interactions. It does not assign an intrinsic interaction depth to an arbitrary fully connected network.

5 Boolean examples and limits of the result

5.1 A fixed noncommutative group

Take $G = A_5$, the even permutations of five objects, and define $f(g) = -1$ exactly when g is a three-cycle. There are 20 three-cycles among the 60 group elements, so $p = 1/3$. Any injection

of A_5 into six Boolean bits gives $A_0 \le \sqrt{6}$. Therefore every network for this predicate with the stipulated maximum-arity-b physical hierarchy satisfies

$$P_{\mathrm{F}} \ge \frac{2}{3\sqrt{3}} \exp\left(\frac{(\log n)^2}{4 \log b} + \frac{\log n}{4} \right).$$

The claim holds for every Boolean extension on the unused four letter codes. It does not prove that an unrestricted network for this word predicate must have the displayed product.

The theorem rules out improving a local multiplication implementation merely by distorting its group-state codes, keeping extra history, or placing more hidden layers inside its local modules. Fixed-seed random-estimator features are also deterministic subword codes and are covered when implemented through the same charged dataflow. Ordinary arithmetic work is not being identified with physical Frobenius product.

5.2 Parity proves that the lower bound is architectural

Take $G = \mathbb{Z}/2\mathbb{Z}$, encode each letter by one sign, and let f be parity. Then $p = 1/2$ and $A_0 = 1$. A binary local physical hierarchy must have $P_{\mathrm{F}} \ge 2^{h(h+1)/4}$ for $n = 2^h$. Yet parity has a shallow tanh realization with polynomial product. Here is a direct construction.

For $x \in \{-1, 1\}^n$, put $s(x) = \sum_i x_i$, $\tau_j = -n + 2j - 1$, and $K = \log(8n)$. Use hidden neurons

$$t_j(x) = \tanh(K(s(x) - \tau_j)), \qquad j = 1, \dots, n,$$

with all shifts supplied by the augmented constant input. Add a constant neuron with value $c_0 = \tanh 1$. If m coordinates are positive, then the ideal signs η_j equal $+1$ for $j \le m$ and -1 otherwise, and

$$1 + \sum_{j=1}^{n} (-1)^j (1 + \eta_j) = (-1)^m.$$

Choose the linear readout

$$z(x) = 2(-1)^n \left[1 + \sum_{j=1}^{n} (-1)^j (1 + t_j(x)) \right],$$

representing its constant term through c_0. Every preactivation before the factor K is an odd integer. Hence $|t_j - \eta_j| \le 2e^{-2K}$, and the output differs from $2 \prod_i x_i$ by at most $4ne^{-2K} < 1$. Its signed margin exceeds one. The first-layer Frobenius norm is $O(Kn^{3/2})$, including the shifts, and the readout norm is $O(\sqrt{n})$. Thus

$$P_{\mathrm{F}} = O(n^2 \log(n + 1)).$$

Its first hidden features access all input letters at once. It is a single global merge, not the prescribed repeated local hierarchy, so there is no contradiction.

5.3 What remains outside the theorem

Nonlocal interactions before a designated merge, globally mixed feature maps, and other organizations of the network are not covered. Neither are uncharged external rereads of input letters at later steps. If such values are physically carried within child boundary states, they are covered as retained history and their carrying matrices must be charged.

Finite ensembles are ordinary deterministic code vectors. The proof also extends to deterministic Hilbert-valued boundary codes and charged Hilbert–Schmidt maps whenever these objects define the

physical realization; the conditional distributions still have finite support over the finite input set. It does not assert a result about arbitrary stochastic networks with unspecified shared randomness.

The coupling lemma can similarly be formulated for finite operations that are bijective in each argument and preserve uniform measure. No additional application is needed here. The group case already shows the limitation of local probabilistic branching-program implementations without introducing internal robustness assumptions.

Finally, no prior distribution occurs in the proof. The Gaussian Bayesian learner in the motivating problem is free to use fully connected global features. An architecture-specific representation lower bound, even one valid for arbitrary internal histories, does not establish a lower bound for its posterior confidence or sample complexity.

Verification and provenance

The conditional-law coupling, physical-band charging, entropy telescope, and constants have conventional independent mathematical checks. This note is not a Lean formalization. Its purpose is to retain a precise obstruction to local compilation methods while leaving the unrestricted representation and Bayesian questions open.